Topic 4: The Short-Time Fourier Transform

Analysis windows, spectrograms, inverse STFT, FFT size and hop size

Digital Signal Processing for Sound and Music · Topic 4

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Lecture roadmap

1 **STFT equation**
local spectra from overlapping frames

2 **Analysis windows**
window shape and spectral leakage

3 **M, N, H**
window size, FFT size, and hop size

4 **Spectrograms**
time-frequency compromise, magnitude and phase

5 **Inverse STFT**
frame synthesis and overlap-add

6 **Example + Exercise 4**
stft-examples.ipynb and E4-STFT

Short-Time Fourier Transform equation

$$X_\ell[k] = \sum_{n=-M/2}^{M/2-1} x\,[n + \ell H] w[n] e^{-j2\pi kn/N}$$

$$k = -N/2, \ldots, N/2 - 1 \qquad \ell = 0,1, \ldots$$

The STFT is not a new transform: it is a sequence of DFTs, each applied to a short windowed frame of the signal.

Symbols

w[n]: analysis window
ℓ: frame index
H: hop size
M: window length; N: FFT size

Interpretation

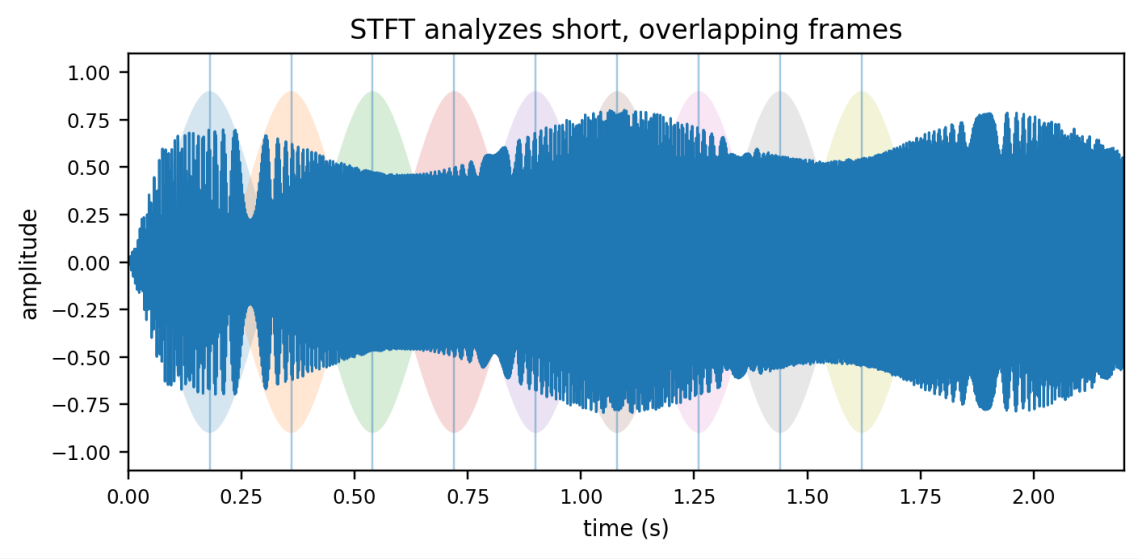
For each frame ℓ, take a local segment centered around time ℓH, multiply it by the window, then compute one DFT column.

Phase convention

Centered indexing around n = 0 makes phase plots easier to interpret and avoids artificial phase ramps.

Why short-time analysis?

- Most musical sounds are not stationary: their spectrum changes over time.
- The STFT assumes the signal is locally stable inside a short frame.
- Frames overlap so changes can be tracked smoothly.



Transform of a windowed sinewave

$$x_w[n] = w[n]A_0\cos(2\pi k_0 n/N)$$
$$= \frac{A_0}{2}w[n]e^{j2\pi k_0 n/N} + \frac{A_0}{2}w[n]e^{-j2\pi k_0 n/N}$$

$$X_w[k] = \frac{A_0}{2}W[k-k_0] + \frac{A_0}{2}W[k+k_0]$$

The phrase “spectrum of a frame” always means spectrum of the windowed frame, not of the raw signal segment alone.

Main idea

A sinusoid multiplied by a window does not produce an ideal impulse in frequency.

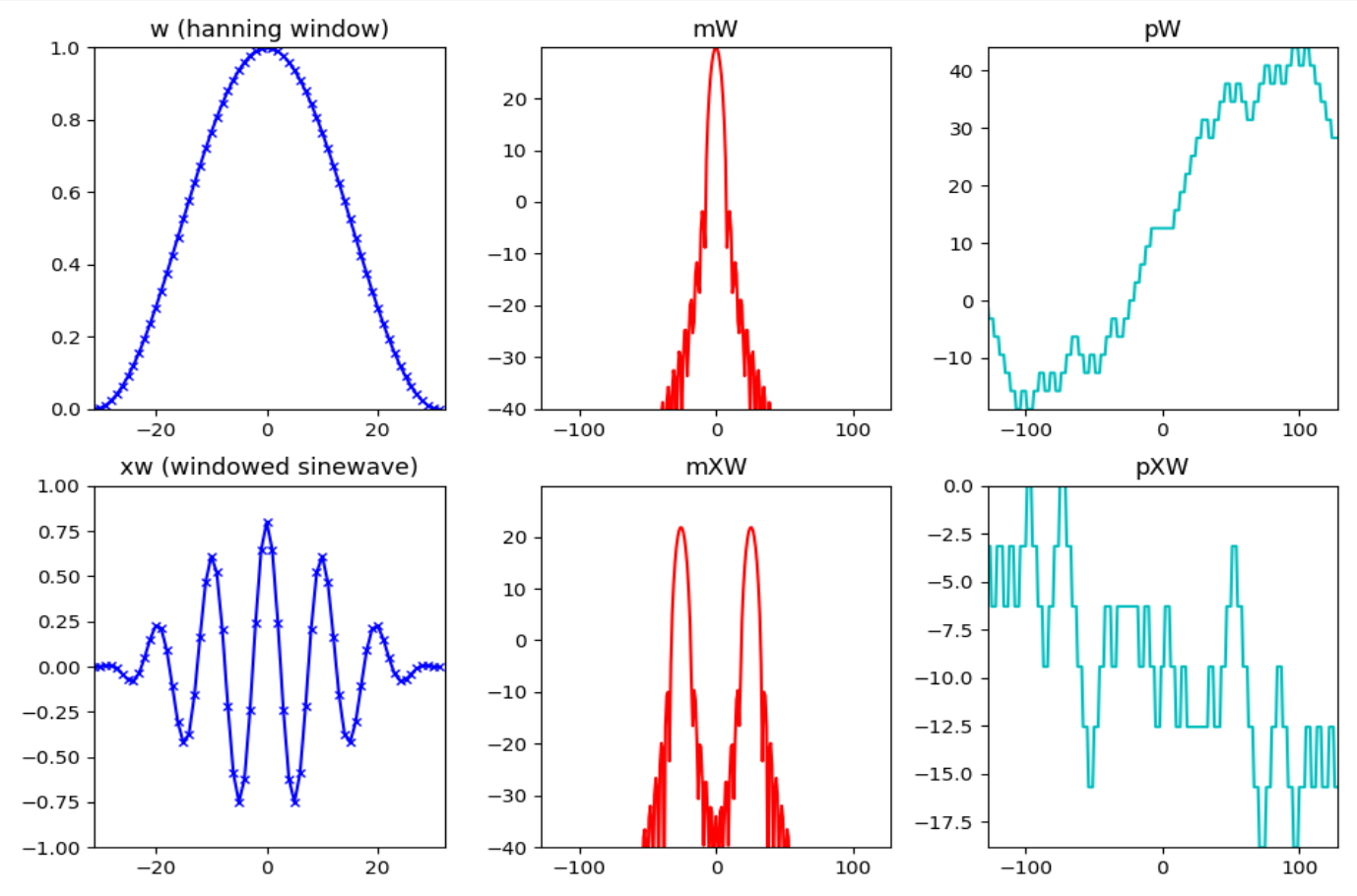
What appears in frequency

The spectrum is two shifted copies of the window spectrum W[k], one at $+k_0$ and one at $-k_0$.

Consequence

The window controls leakage and peak shape: main-lobe width and side-lobe level.

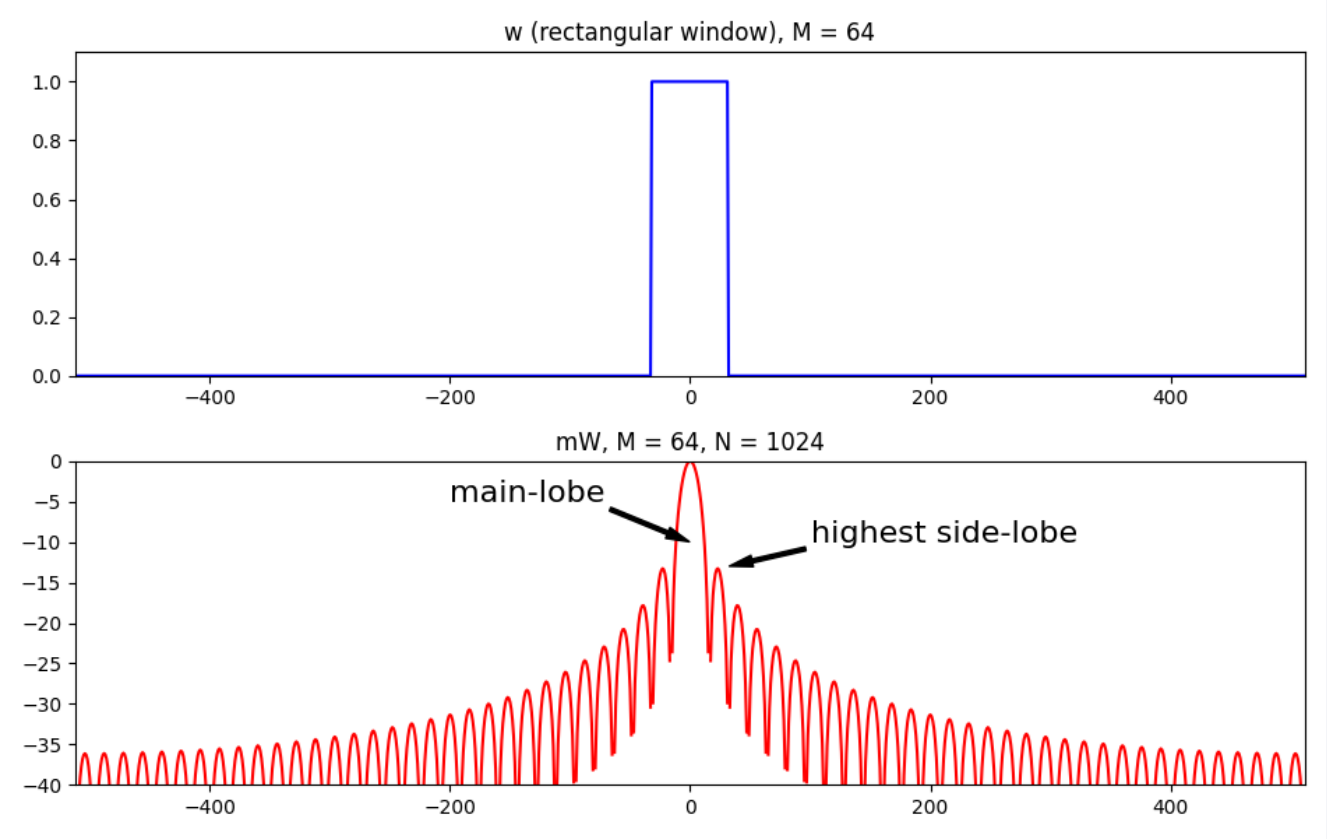
Windowed sinewave: what the formula looks like



- The time signal is a sinusoid tapered by the analysis window.
- The magnitude spectrum has peaks around $\pm k_0$, shaped by W[k].
- The phase is meaningful mainly where magnitude is significant.

This is why magnitude-only plots can hide important assumptions about windowing and phase.

Analysis window



Window shape

A window reduces discontinuities at frame boundaries, but it also has its own spectrum.

Main lobe

Controls frequency resolution: narrow lobes separate close sinusoids better.

Side lobes

Control leakage: lower side lobes reduce interference from stronger nearby components.

Window functions in SciPy

Common scipy.signal window functions used in STFT analysis

boxcar / rectangular	maximum resolution, high leakage	**flattop**	amplitude accuracy
bartlett	triangular window	**gaussian**	Gaussian shape, parameter std
barthann	modified Bartlett-Hann	**general_gaussian**	generalized Gaussian shape
hann / hanning	good general-purpose taper	**kaiser**	tunable beta parameter
hamming	lower first side-lobe	**chebwin**	tunable attenuation
blackman	stronger side-lobe suppression	**parzen**	Parzen window
blackmanharris	very low side-lobes	**triang**	triangular window
bohman	Bohman window	**slepian / dpss**	energy concentration
nuttall	minimum 4-term Blackman-Harris		

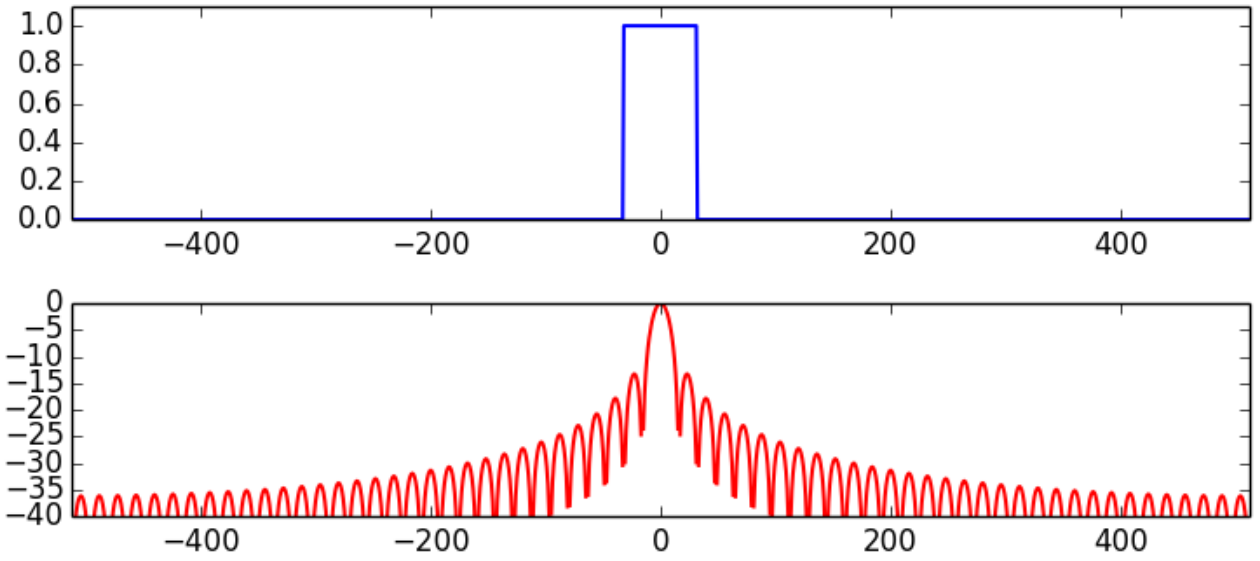
In code: `scipy.signal.get_window(name, M)`. Always report the window name together with M, N, and H.

Rectangular window

$$w[n] = 1, \quad n = -\frac{M}{2}, \ldots, \frac{M}{2}$$

$$W[k] = \frac{\sin(\pi k)}{\sin(\pi k / M)}$$

- main-lobe width: 2 bins
- side-lobe level: −13.3 dB
- Best frequency resolution but poor leakage control.

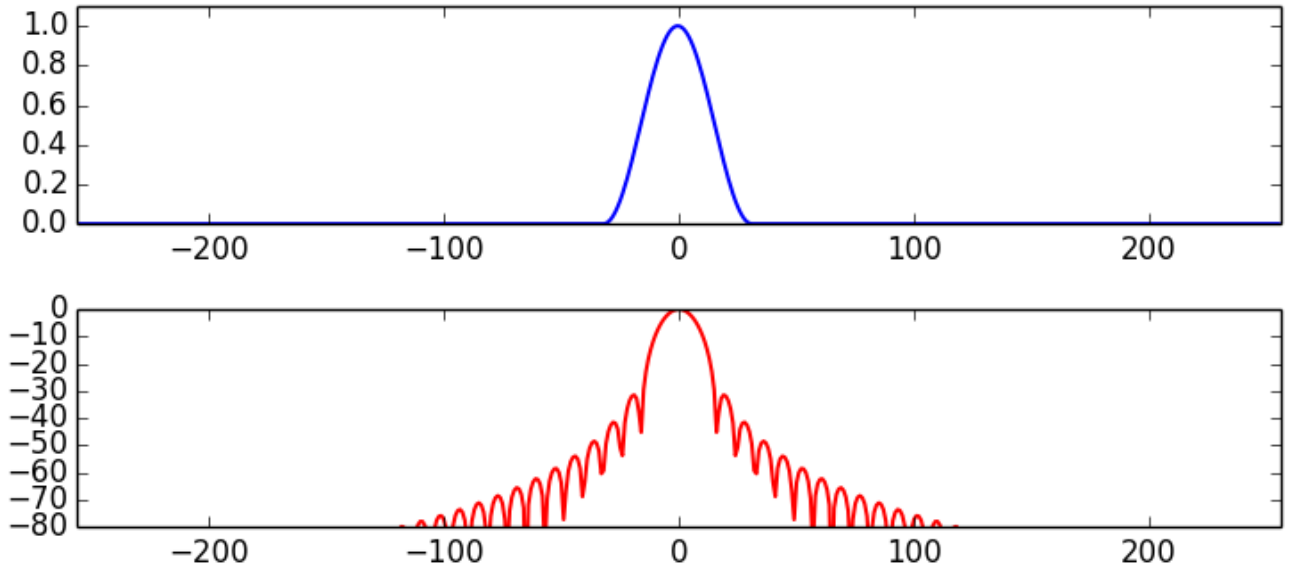


Hann window

$$w[n] = 0.5 + 0.5\cos(2\pi n/M)$$

$$W[k] = 0.5D[k] + 0.25(D[k-1] + D[k+1])$$

- main-lobe width: 4 bins
- side-lobe level: −31.5 dB
- A common default for spectrograms.

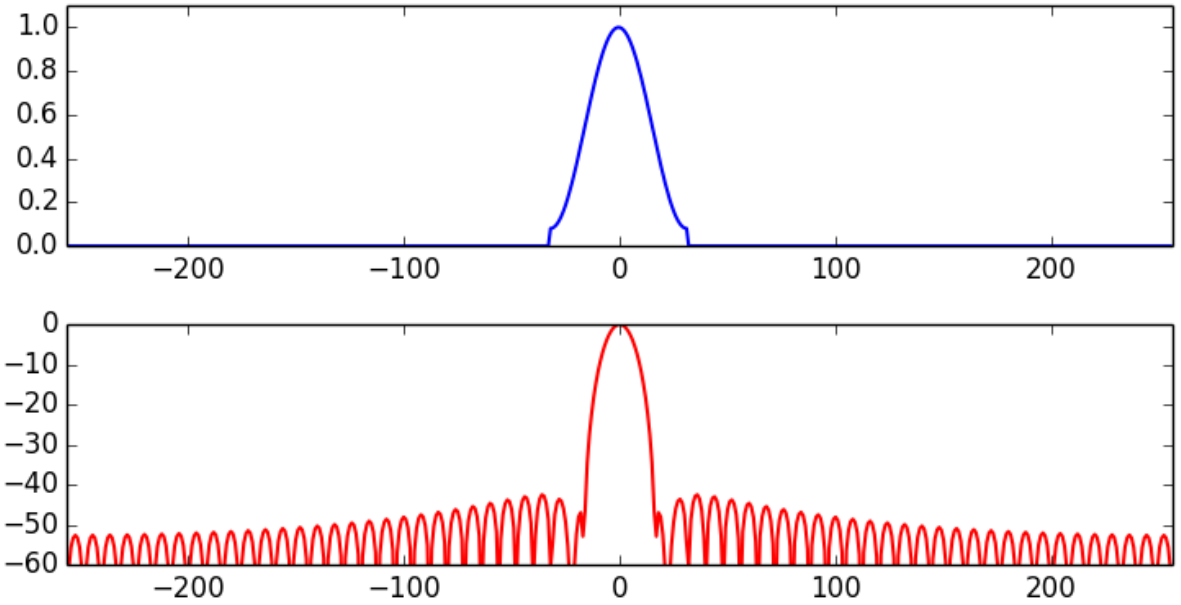


Hamming window

$$w[n] = 0.54 + 0.46\cos(2\pi n/M)$$

$$W[k] = 0.54D[k] + 0.23(D[k-1] + D[k+1])$$

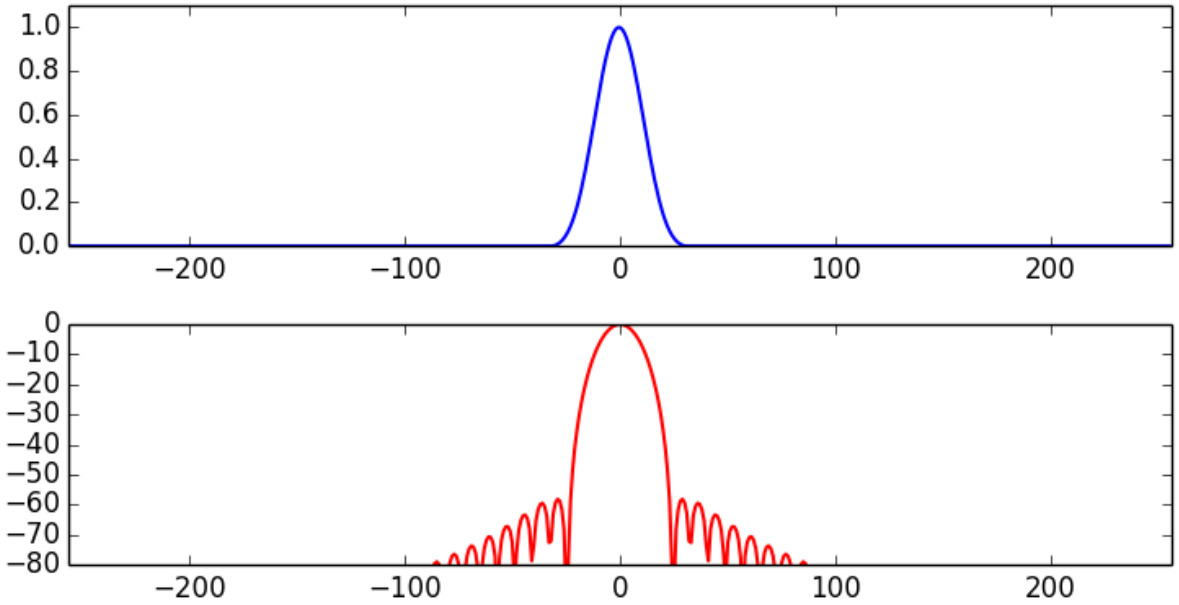
- main-lobe width: 4 bins
- side-lobe level: −42.7 dB
- Lower first side-lobe than Hann.



Blackman window

$$w[n] = 0.42 - 0.5\cos(2\pi n/M) + 0.08\cos(4\pi n/M)$$

- main-lobe width: 6 bins
- side-lobe level: −58 dB
- Good leakage suppression, wider peaks.



Blackman-Harris window

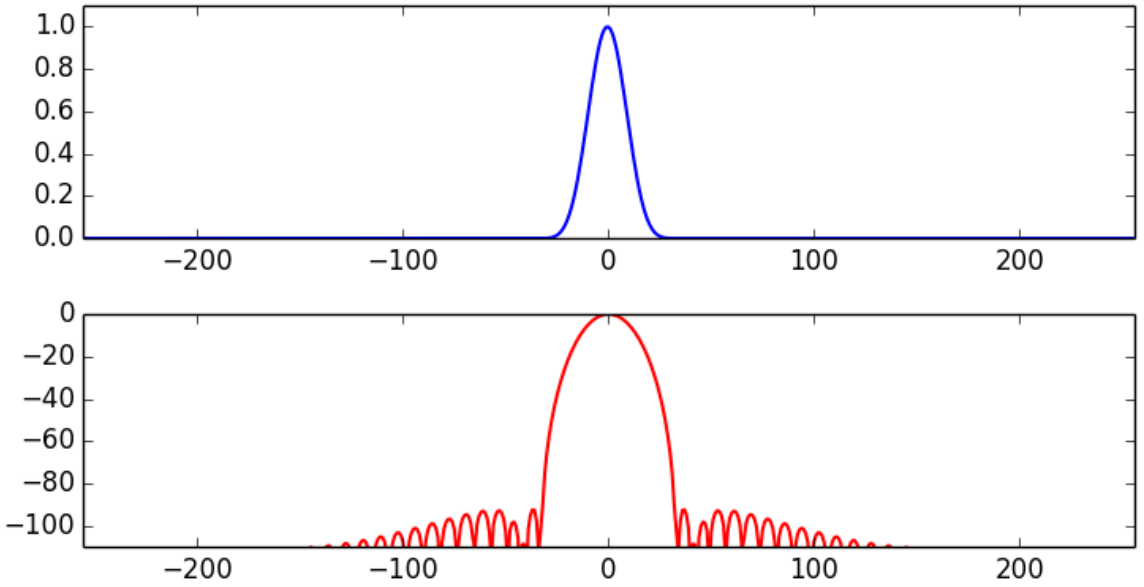
$$w[n] = a_0 - a_1\cos\left(\frac{2\pi n}{M-1}\right) + a_2\cos\left(\frac{4\pi n}{M-1}\right) - a_3\cos\left(\frac{6\pi n}{M-1}\right)$$

$a_0 = 0.35875, \quad a_1 = 0.48829, \quad a_2 = 0.14128, \quad a_3 = 0.011$

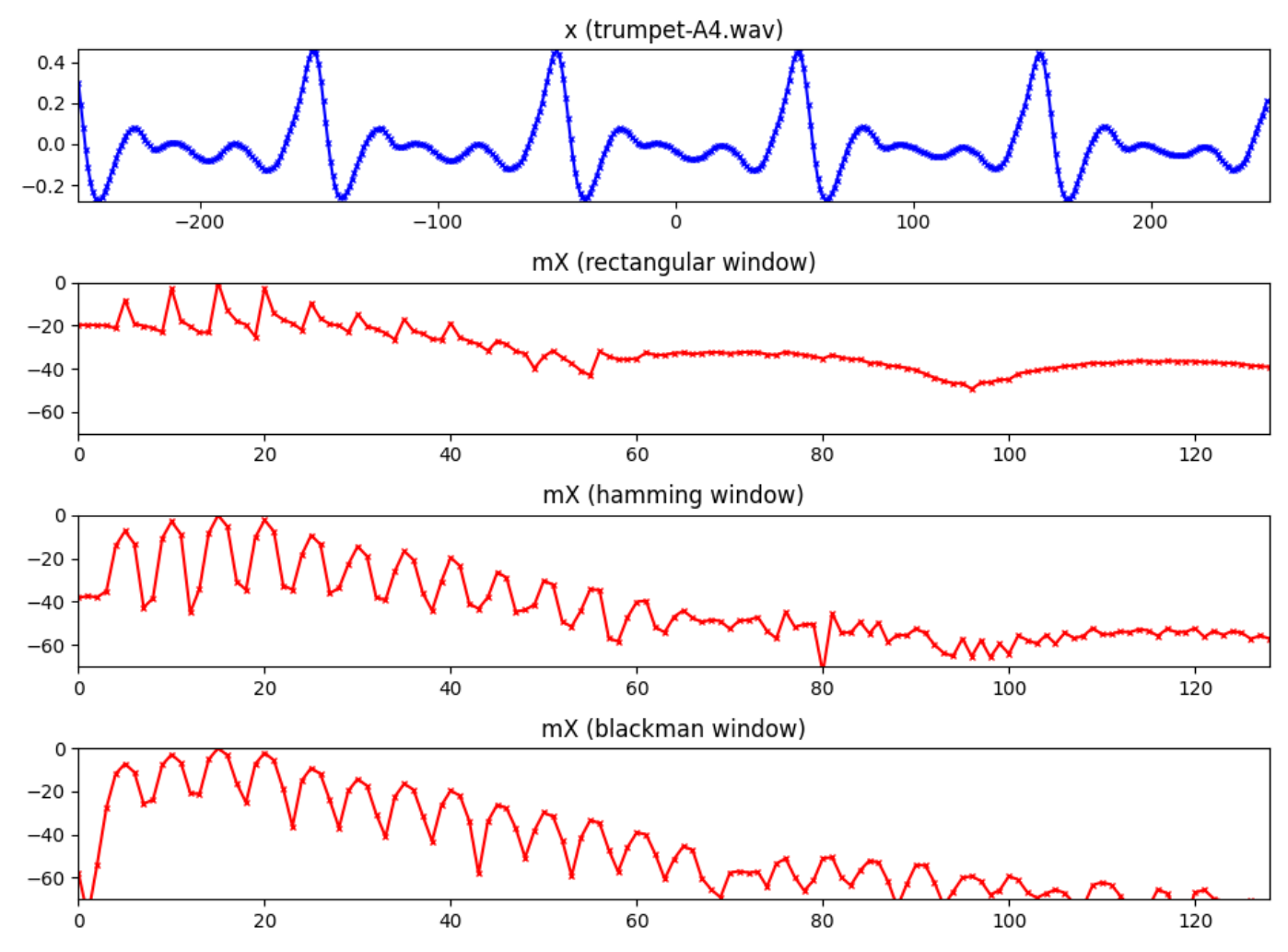
- main-lobe width: 8 bins
- side-lobe level: about −92 dB
- Excellent side-lobe suppression, but lower frequency resolution.

Interpretation

Use it when weak components near strong components matter more than resolving very close frequencies.



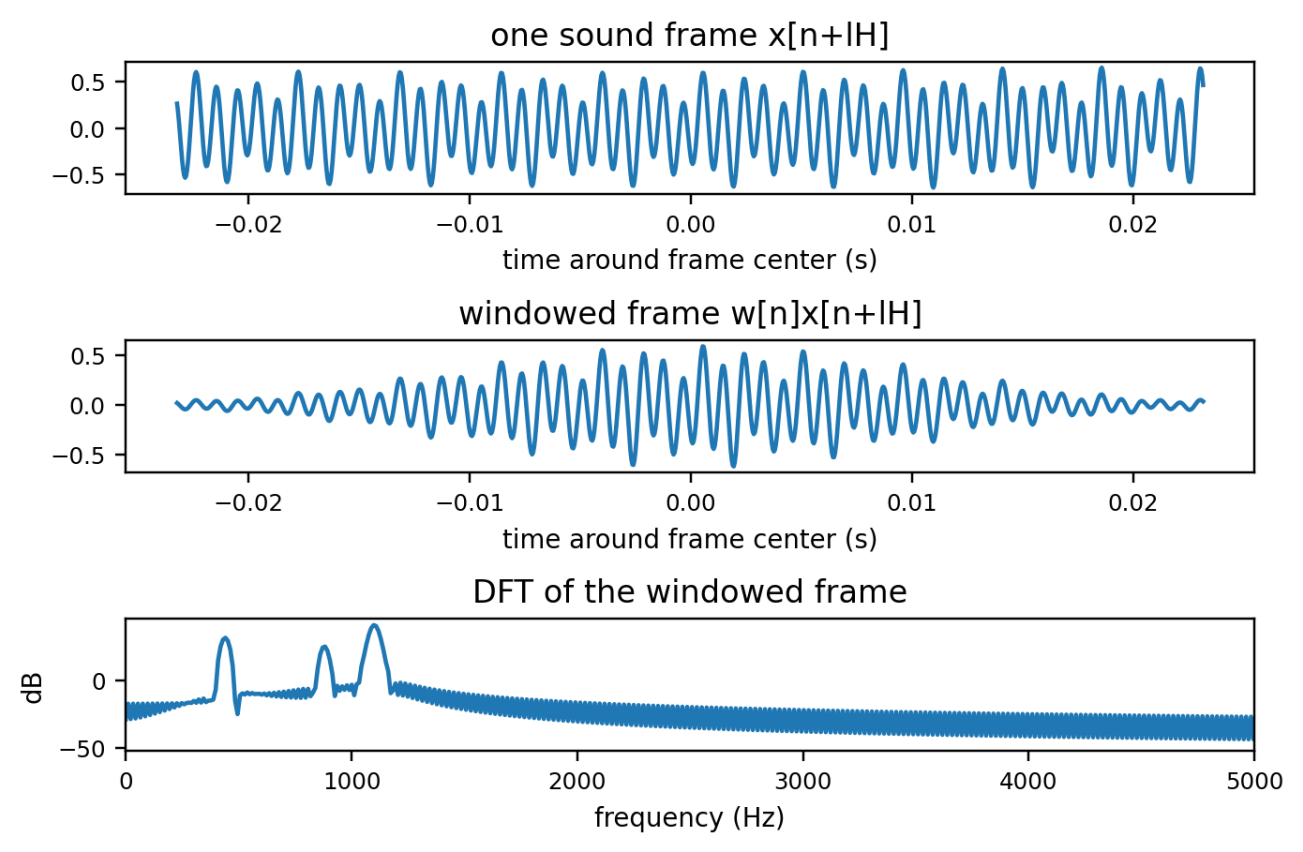
Window choice changes the spectrum



- The same frame produces different spectra depending on the window.
- Rectangular windows show strong side-lobes.
- Hamming and Blackman reduce leakage but widen peaks.
- No window is best for every task.

Always interpret a spectral plot together with the window that produced it.

STFT and analysis window



Local analysis

The STFT converts each short windowed fragment into a complex spectrum.

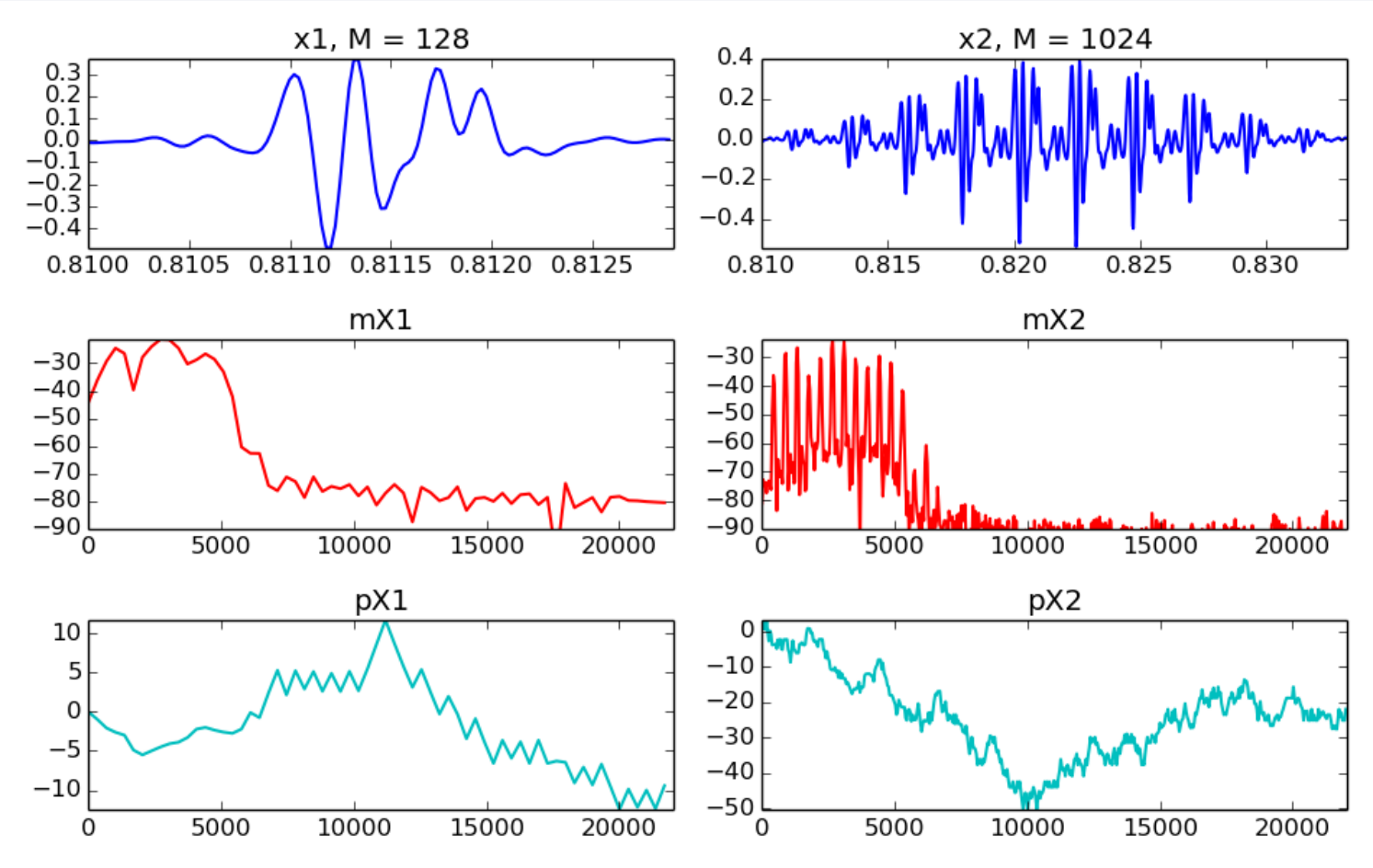
What changes from frame to frame?

The frame center lH moves through the sound; each new frame produces a new column in the spectrogram.

What remains fixed?

The analysis parameters: window type, M, N, and H.

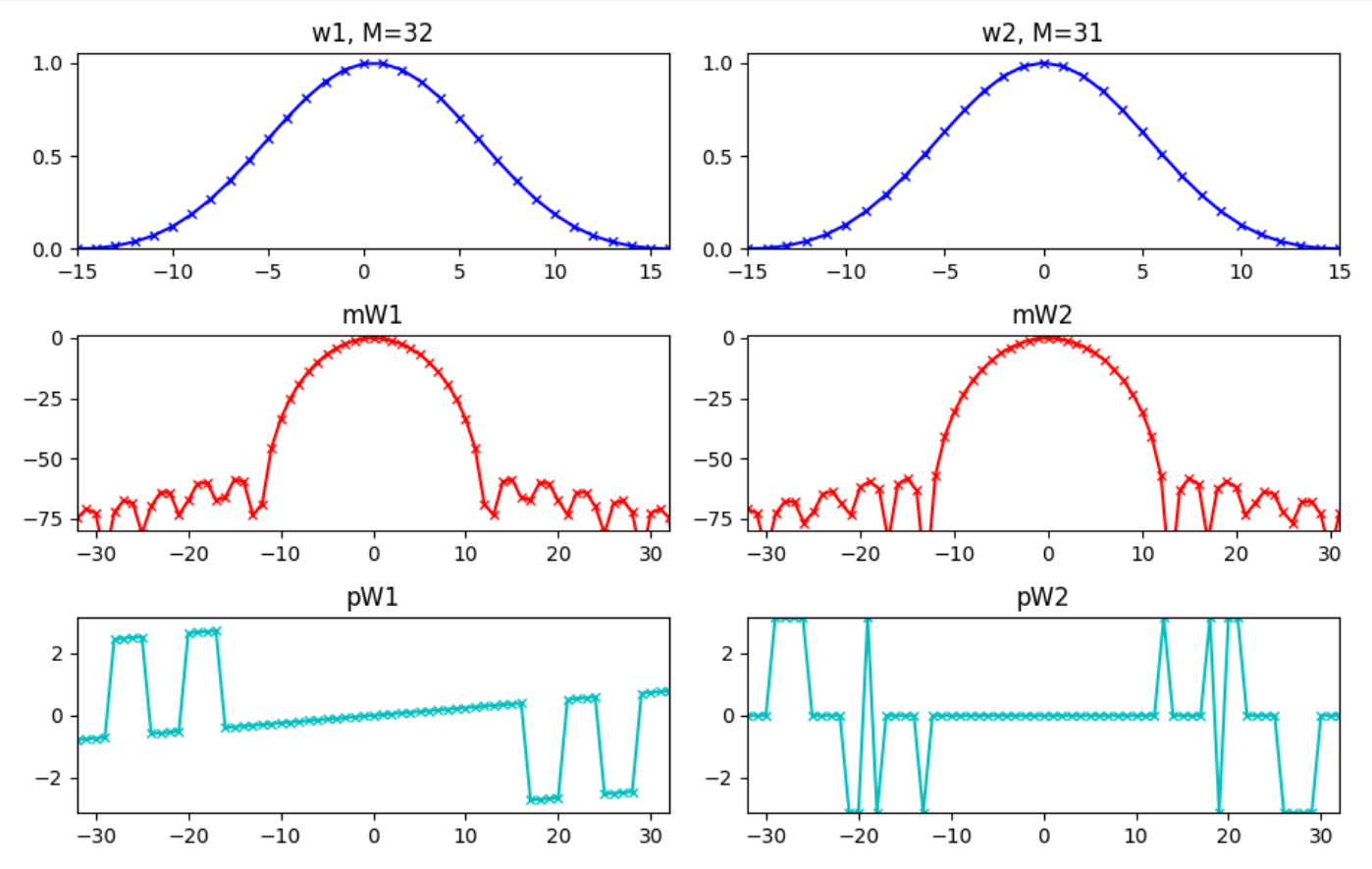
Window size



- Small M: better time localization, poorer frequency discrimination.
- Large M: better frequency discrimination, poorer time localization.
- The “right” M depends on how fast the sound changes and which frequencies matter.

This is the central time-frequency compromise of the STFT.

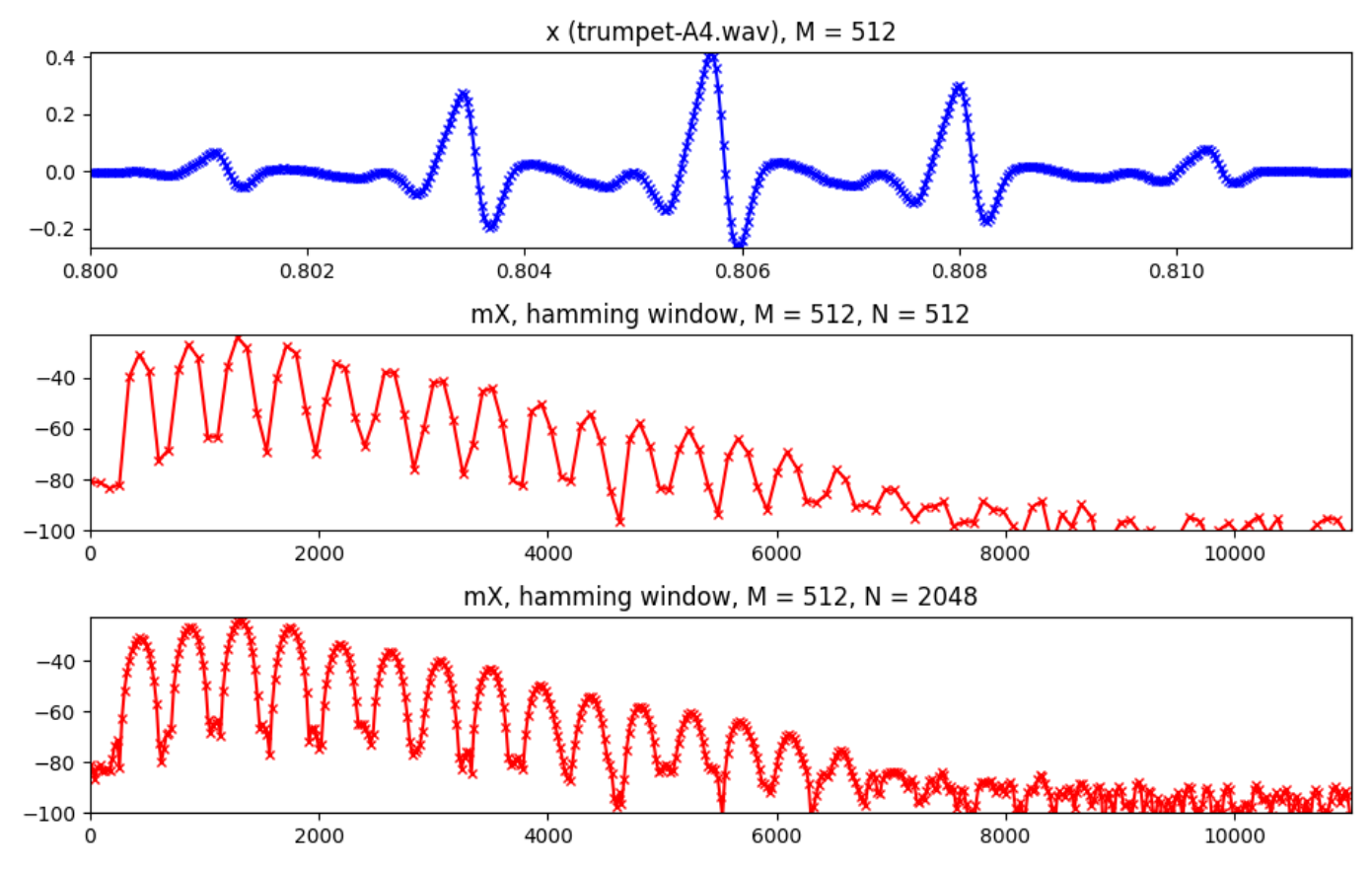
Even-odd size window



- The window length M can be even or odd.
- Centering conventions affect how samples are arranged around n = 0.
- For phase-sensitive analysis, be explicit about indexing and zero-phase placement.

In implementation, a one-sample mismatch can introduce a visible phase difference.

FFT size



- N controls the number of DFT samples used to represent the spectrum.
- When N > M, the frame is zero-padded before the FFT.
- Zero-padding interpolates the spectrum; it does not add new information.

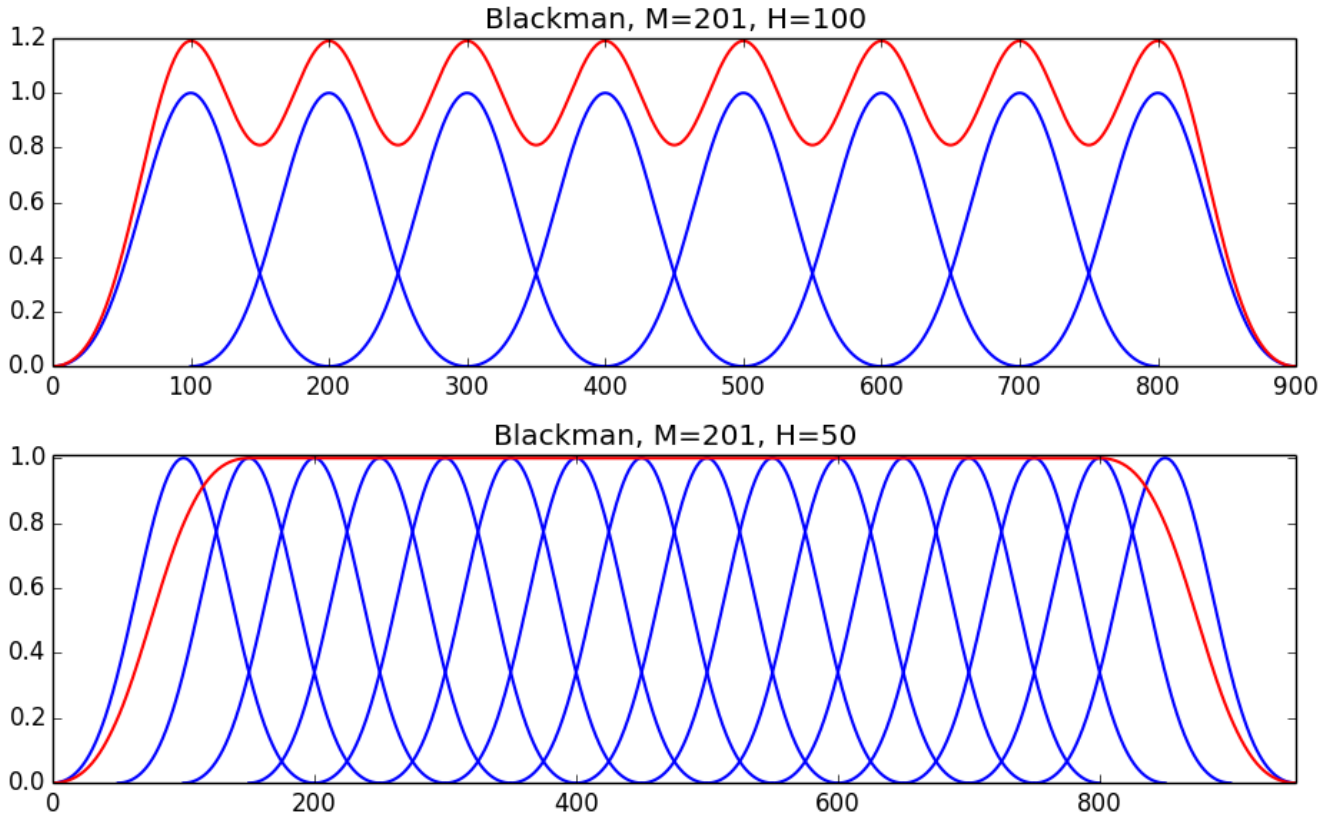
Rule of thumb

M controls analysis-window duration; N controls how densely the spectrum is sampled.

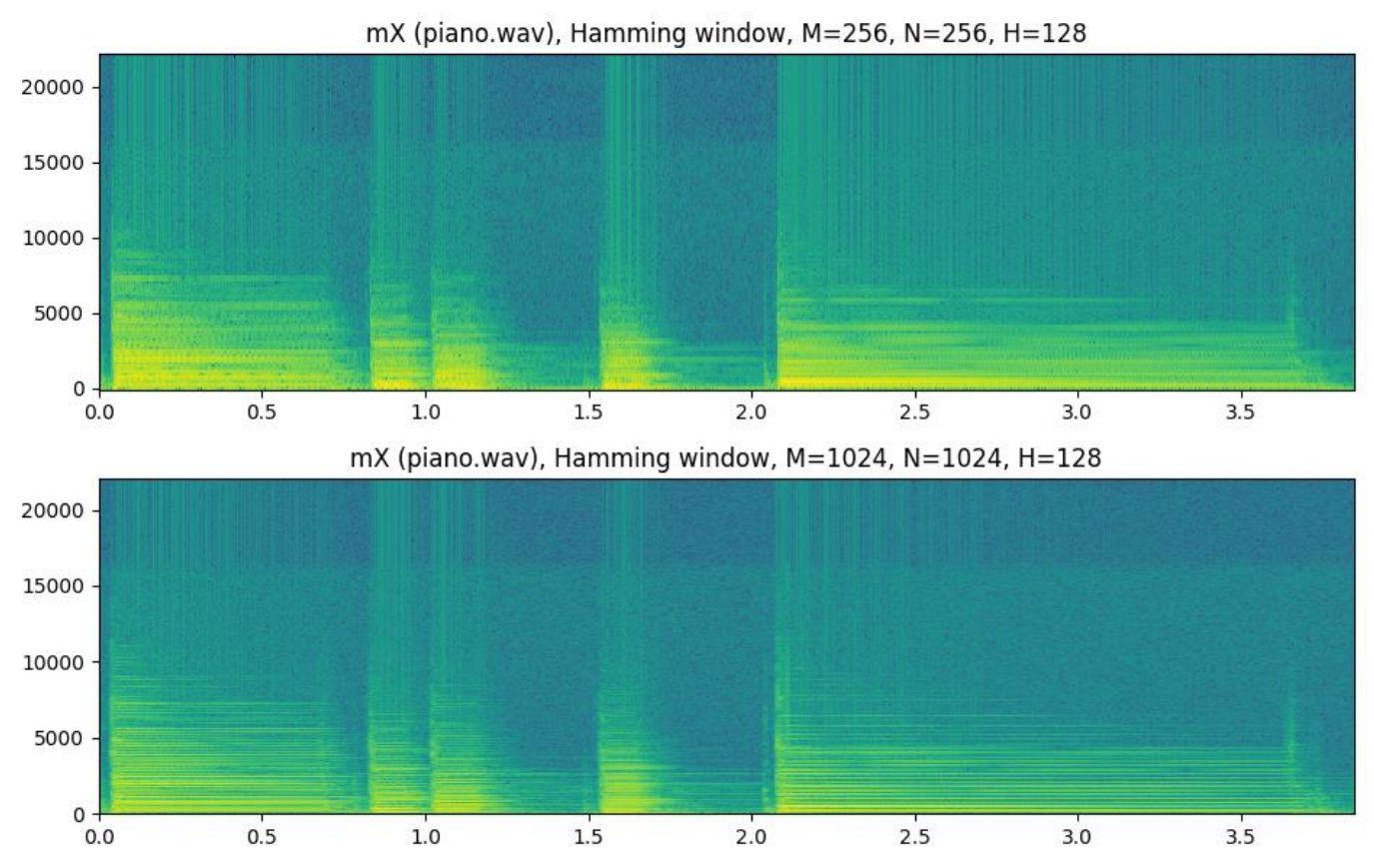
Hop size

$$A_w[n] = \sum_{\ell=0}^{L-1} w\,[n - \ell H] = c$$

- H is the distance in samples between consecutive frame centers.
- Smaller H means more overlap and smoother time tracking.
- For good resynthesis, overlapped windows should sum to a near-constant value.



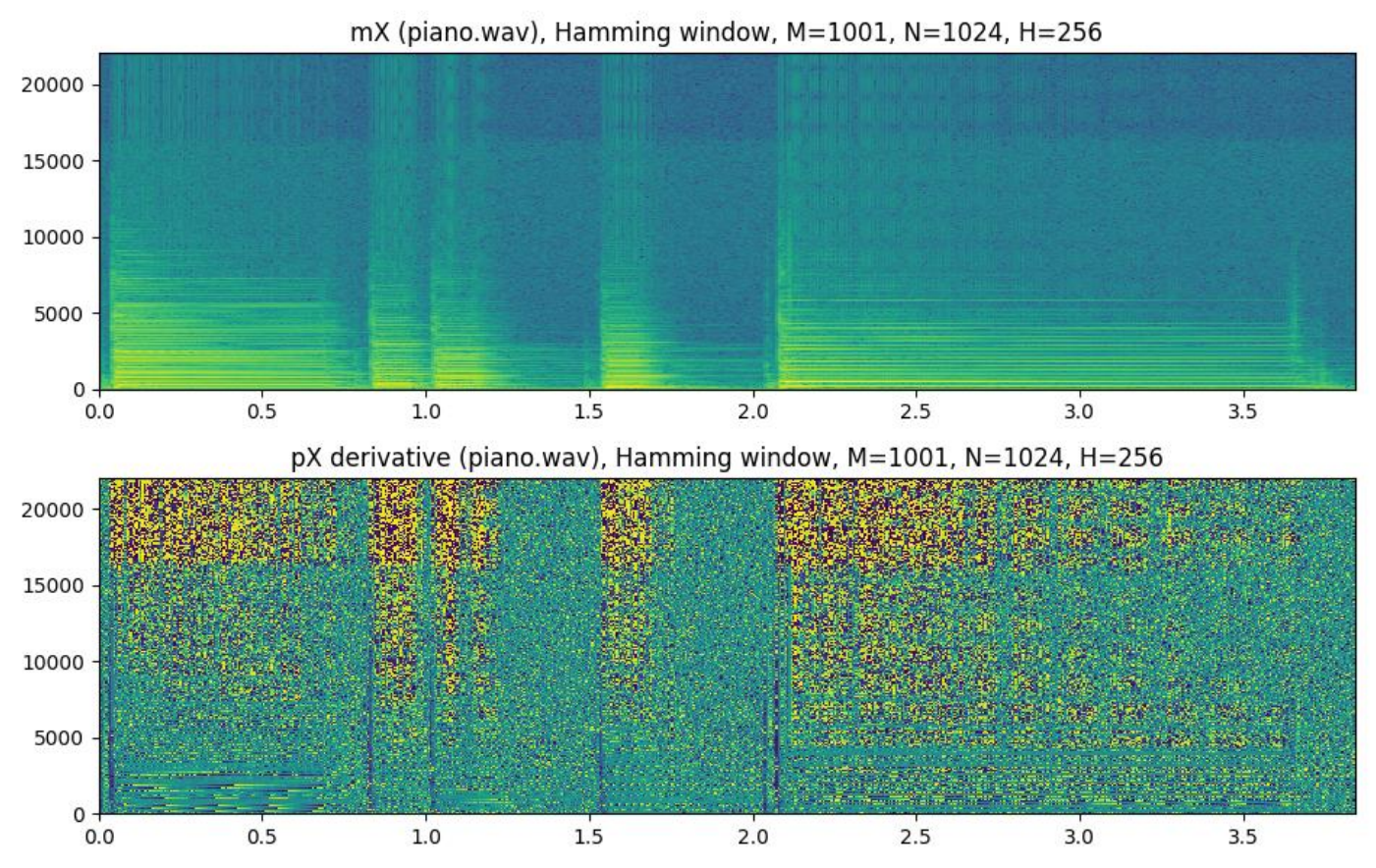
Time-frequency compromise



- Short windows follow transients and rapid changes.
- Long windows distinguish close frequencies more clearly.
- The spectrogram is never “the signal”; it is a view determined by M, N, H, and the window.

When comparing spectrograms, always compare analysis parameters first.

Amplitude and phase spectrograms



- Magnitude spectrograms show energy distribution over time and frequency.
- Phase spectrograms store temporal alignment information.
- Phase often looks noisy when plotted directly, but it is essential for reconstruction and transformations.

Do not interpret phase where magnitude is close to zero.

Inverse STFT

$$\hat{x}_\ell[n] = \frac{1}{N} \sum_{k=-N/2}^{N/2-1} X_\ell[k] e^{j2\pi kn/N}$$

$$\tilde{x}_\ell[n] = \hat{x}_\ell[n - \ell H] w_s[n - \ell H]$$

$$y[n] = \frac{\sum_\ell \tilde{x}_\ell[n]}{\sum_\ell w[n - \ell H] w_s[n - \ell H]}$$

Implementation logic: IFFT each frame → place it in time → overlap-add → compensate the window sum.

1 · Inverse DFT

Convert each spectral column Xℓ[k] back into a time-domain frame.

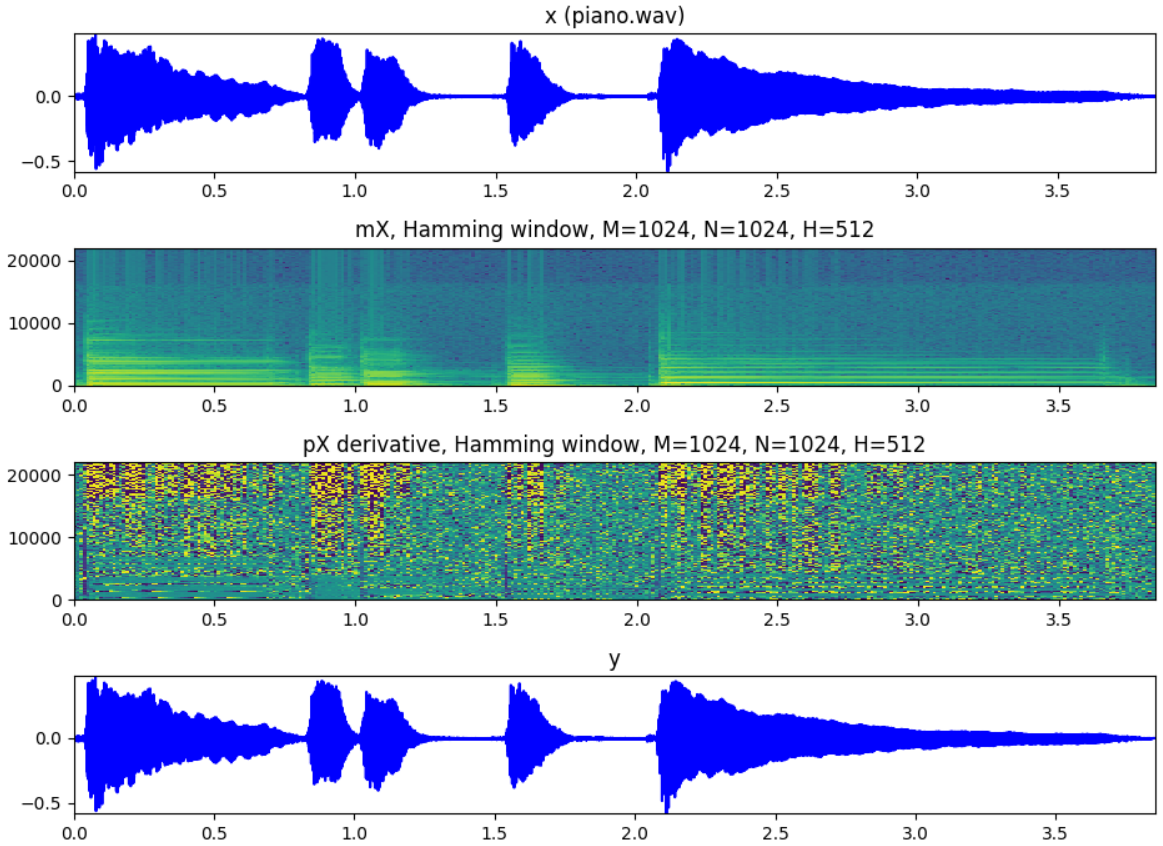
2 · Window + shift

Apply the synthesis window, move the frame to time ℓH, and add it to the output buffer.

3 · Normalize

Divide by the overlap of analysis/synthesis windows. With the same window this becomes a sum of w^2 terms.

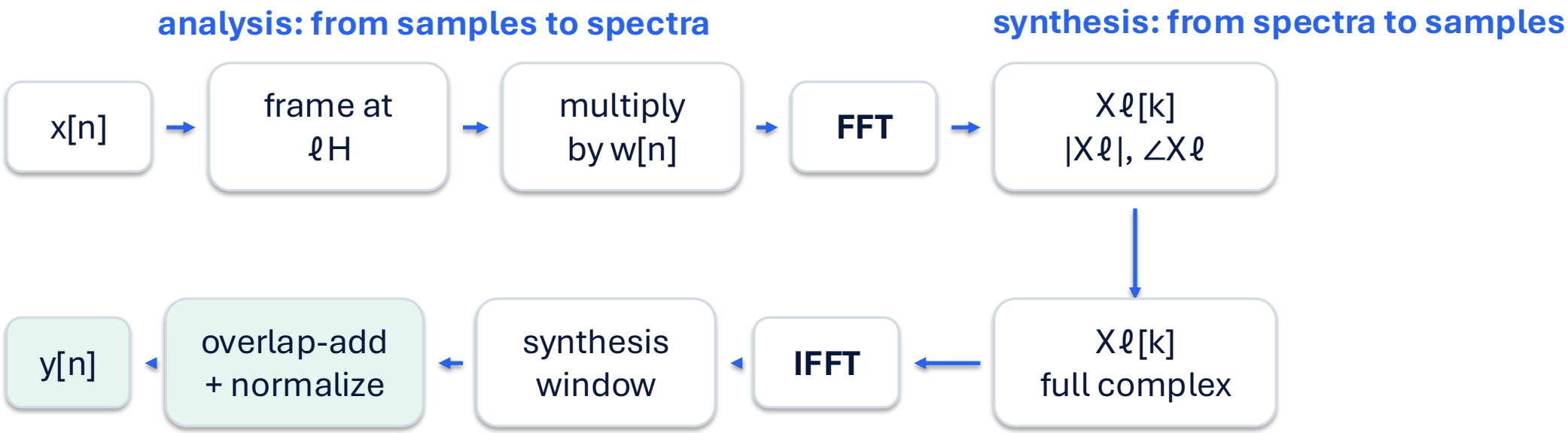
Analysis/synthesis with magnitude and phase



$$X_\ell[k] \rightarrow (mX_\ell[k],\ pX_\ell[k]) \rightarrow y[n]$$

- The complex STFT contains magnitude and phase for every frame.
- IFFT reconstructs each frame.
- Overlap-add reconstructs the full time signal.
- Magnitude-only reconstruction is generally not exact.

STFT system



Parameters

window type · M · N · H

Stored data

magnitude and phase for every frame

Check

compare x[n] and y[n], then listen

Analysis creates spectra on a frame grid; synthesis uses the same grid. Exact reconstruction needs phase plus proper overlap-add normalization.

Implementation · stftAnal()

How sms-tools converts overlapping waveform frames into magnitude and phase spectra

```
mX, pX = STFT.stftAnal(x, w, N, H)
```

01 Pad signal
M/2 zeros
at both ends

→

02 Center frames
length M;
advance by H

→

03 Normalize w
w / sum(w)

→

04 DFT.dftAnal()
window +
zero-phase FFT

→

05 Store spectra
one row per
analysis frame

Inputs & parameters

x: audio samples w: analysis window (M samples)
N: FFT size (power of 2; N ≥ M) H: hop size
Frames are analyzed at positions separated by H.

Output arrays

mX: magnitude [dB] pX: phase [radians]
Both arrays: frames × (N/2 + 1) frequency bins
Phase is unwrapped within each frame by dftAnal().

Source: github.com/MTG/sms-tools · smstools/models/stft.py

Implementation · stftSynth()

How sms-tools turns magnitude and phase spectra into a time-domain signal

```
y = STFT.stftSynth(mX, pX, M, H)
```

1 Read a frame

mX[i, :], pX[i, :]

→

2 DFT.dftSynth()

convert dB + phase
to M samples

→

3 Scale by H

H × reconstructed
frame

→

4 Overlap-add

shift by H; trim
edge padding

What the function expects

mX and pX: same matrices returned by stftAnal()
M: original analysis-window length
H: same hop size used in the analysis

Reconstruction details

N is inferred from the spectrum width, not passed.
The code uses H × frame in overlap-add and
then removes the padded edges of the output.

Source: github.com/MTG/sms-tools · smstools/models/stft.py

Example notebook: STFT workflow

`examples/stft-examples.ipynb` connects the STFT equations to a complete experiment: analyze, inspect, synthesize, and listen.

1 Load sound	2 Choose window,M,N,H	3 STFT analysis	4 Inspect mX,pX	5 IFFT + OLA	6 Listen check
wavread select sound	M window N FFT · H hop	frame window + FFT	magnitude phase	inverse FFT overlap-add	compare audio

Parameter experiments

Change one value at a time:
- M: time/frequency trade-off
- N: spectral sampling / zero-padding
- H: overlap and reconstruction smoothness

Spectrogram reading

mX shows energy over time/frequency. pX stores phase; interpret it mainly where magnitude is significant.

Resynthesis check

Use both magnitude and phase. Compare waveform error and listen for discontinuities, smearing, or modulation artifacts.

Reading strategy: every plot should be tied to the exact windowed frame, centering convention, FFT size, and hop size that produced it.

How to read the STFT example outputs

1 · Windowing

The spectrum belongs to the windowed frame. Changing the window changes leakage and peak shape.

2 · Frame and hop

M determines how much sound each spectrum sees; H determines how often spectra are computed.

3 · Magnitude and phase

Magnitude makes the time-frequency structure visible; phase is required for reconstruction.

4 · Symmetry

For real input frames, positive and negative frequencies are conjugate-symmetric before one-sided display.

5 · Resynthesis

The IFFT and overlap-add steps test whether the analysis representation preserves the signal.

6 · Parameter sensitivity

Change M, N, H and window type one at a time; listen and inspect the result.

Exercise 4: STFT analysis and onset detection

Four connected tasks: characterize windows, verify reconstruction, measure band energy and detect attacks.

01 Window main lobe

Find the first minima around the central peak.
Compare boxcar, Hamming and Blackman–Harris.

02 Reconstruction and SNR

Run STFT → inverse STFT. Compare the SNR of the full signal and its interior.

03 Frequency-band energy

Convert mX (dB) to linear magnitudes; sum $|X|^2$ in the 0–3 kHz and 3–10 kHz bands.

04 Onset detection function

Compute increases between consecutive dB energy frames; discard negative differences.

THE SIGNAL-PROCESSING CHAIN



Notebook: E4-STFT.ipynb • Compare piano.wav (short/long windows) and sax-phrase-short.wav.

Onset detection: positive changes in band energy

A new note or acoustic event often causes a sudden energy increase between consecutive STFT frames.

COMPUTATION FOR EACH FRAME ℓ

1 Estimate energy separately in two bands

$$|X_\ell[k]| = 10^{(mX_\ell[k]/20)}$$

$$E_\beta(\ell) = 10 \log_{10}\left(\sum_{k \in \beta} |X_\ell[k]|^2 \right)$$

Low: $0 < f < 3000$ Hz High: $3000 < f < 10000$ Hz

2 Subtract the preceding energy frame

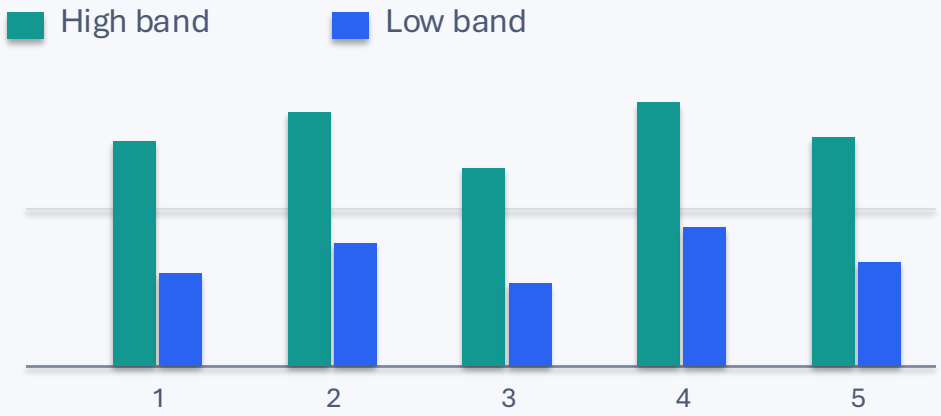
$$O_\beta(\ell) = E_\beta(\ell) - E_\beta(\ell-1) \qquad O_\beta(0) = 0$$

3 Retain rises, suppress energy decreases

$$ODF_\beta(\ell) = \max(O_\beta(\ell), 0)$$

Half-wave rectification keeps onsets, not offsets.

INTERPRETING THE ODF



Illustrative peaks at five successive note attacks

Piano lab (M = 513): expect five high-band peaks above 10 dB.

Compare M = 2047: longer windows smear attacks. A single fixed threshold may not suit every sound.

What to remember from Topic 4

1 **STFT**
a sequence of windowed DFTs

2 **Window**
controls leakage and time-domain framing

3 **M**
time/frequency analysis window length

4 **N**
FFT size and spectral sampling density

5 **H**
hop size and overlap-add behavior

6 **ISTFT**
reconstruction needs magnitude, phase, and compatible windows

The STFT is an analysis method, not a neutral picture: every spectrogram reflects the chosen frame, window, FFT size and hop size.

References and credits

- sms-tools models: https://github.com/MTG/sms-tools
- sms-tools-materials lecture slides and plotting code.
  - Example notebook: `examples/stft-examples.ipynb`
  - Exercise notebook: `exercises/E4-STFT.ipynb`
- Wikipedia background: STFT, window function, spectrogram.
- Julius O. Smith, Spectral Audio Signal Processing: STFT and windowing chapters.
- Slides released under CC Attribution-Non Commercial-Share Alike license and code under Affero GPL license.